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ELECTRODE CALENDERING WITH ROLLERS HAVING CONTROLLABLE SLEEVES

Abstract

An electrode calendering system for reducing post-calendering electrode wrinkling includes at least one roller configured to compress an electrode and a sleeve. The roller is defined by an outer surface configured to contact the electrode. The outer surface has a first edge portion at a first side of the roller and a second edge portion at a second side of the roller opposing the first side. The first edge portion and the second edge portion of the roller align with first and second outer edges of the electrode when the electrode contacts the roller. The sleeve is disposed about the outer surface of the roller along the first edge portion, the sleeve having a controllable thickness to adjust a pressure between the first outer edge of the electrode and the sleeve. Other example systems and methods for reducing post-calendering electrode wrinkling are also disclosed.

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Background/Summary

INTRODUCTION

[0001] The information provided in this section is for the purpose of generally presenting the context of the disclosure. Work of the presently named inventors, to the extent it is described in this section, as well as aspects of the description that may not otherwise qualify as prior art at the time of filing, are neither expressly nor impliedly admitted as prior art against the present disclosure.

[0002] The present disclosure relates to electrode calendering with rollers having controllable sleeves, and more particularly to electrode calendering with rollers having sleeves with thicknesses that are actively and/or passively adjusted to reduce wrinkling of electrodes.

[0003] Electric power sources, such as batteries for vehicles include anode and cathode electrodes. An anode electrode may include lithium. In such examples, a lithium metal anode can be formed by applying a layer of lithium material to a layer of copper material to form a stack and then cold rolling (or calendering) the stack between a pair of rollers.

SUMMARY

[0004] An electrode calendering system for reducing post-calendering electrode wrinkling includes at least one roller configured to compress an electrode and a sleeve. The roller is defined by an outer surface configured to contact the electrode. The outer surface has a first edge portion at a first side of the roller and a second edge portion at a second side of the roller opposing the first side. The first edge portion and the second edge portion of the roller align with first and second outer edges of the electrode when the electrode contacts the roller. The sleeve is disposed about the outer surface of the roller along the first edge portion, the sleeve having a controllable thickness to adjust a pressure between the first outer edge of the electrode and the sleeve.

[0005] In other features, the sleeve is a first sleeve, the electrode calendering system further includes a second sleeve disposed on the outer surface of the roller along the second edge portion, and the second sleeve has a controllable thickness to adjust a pressure between the second outer edge of the electrode and the second sleeve.

[0006] In other features, the thickness of the sleeve adjusts passively to modulate the pressure between the first outer edge of the electrode and the sleeve.

[0007] In other features, the sleeve includes a plurality of knitted courses each having loops interlocking with loops of an adjacent knitted band of the plurality of knitted bands.

[0008] In other features, one or more of the knitted courses includes an inlay element woven therein.

[0009] In other features, the sleeve includes multiple inlaid elements having a sinusoidal shape in a circumferential direction relative to the roller.

[0010] In other features, the plurality of knitted courses includes a first set of courses formed of a smart material and a second set of courses formed of non-smart materials, and the first set of courses and the second set of courses are collocated or alternated in occurrence within the sleeve.

[0011] In other features, the sleeve includes a smart material element.

[0012] In other features, the smart material element includes a super-elastic shape memory alloy.

[0013] In other features, the sleeve includes a support layer and the smart material element integrated into the support layer.

[0014] In other features, the support layer is a first support layer, and the sleeve includes a second support layer disposed along a surface of the first support layer such that a portion of the layer of

the smart material is positioned between the first support layer and the second support layer.

[0015] In other features, the electrode calendering system further includes at least one optical sensor disposed downstream of the roller, the control module in communication with the optical sensor, and the control module is configured to receive a signal from the optical sensor and control the temperature control device based on the signal from the optical sensor to adjust a temperature of the sleeve, thereby adjusting the pressure between the first outer edge of the electrode and the sleeve.

[0016] In other features, the thickness of the sleeve is actively controlled to adjust the pressure between the first outer edge of the electrode and the sleeve.

[0017] In other features, the electrode calendering system further includes a force or pressure sensor disposed between the sleeve and the roller, a temperature control device adjacent to the sleeve, and a control module in communication with the force or pressure sensor and the temperature control device, the control module configured to receive a signal from the force or pressure sensor and control the temperature control device based on the signal to adjust a temperature of the sleeve, thereby adjusting the pressure between the first outer edge of the electrode and the sleeve.

[0018] An electrode calendering method for reducing post-calendering electrode wrinkling, includes receiving an electrode at a roller. The roller is defined by an outer surface configured to contact the electrode. The outer surface has a first edge portion at a first side of the roller and a second edge portion at a second side of the roller opposing the first side. The first edge portion and the second edge portion of the roller align with first and second outer edges of the electrode when the electrode contacts the roller. The electrode calendering method further includes controlling a thickness of a sleeve disposed about the outer surface of the roller along the first edge portion, to adjust a pressure between the first outer edge of the electrode and the sleeve.

[0019] In other features, controlling the thickness of the sleeve includes passively adjusting the thickness of the sleeve to modulate the pressure between the first outer edge of the electrode and the sleeve.

[0020] In other features, controlling the thickness of the sleeve includes actively controlling thickness of the sleeve to adjust the pressure between the first outer edge of the electrode and the sleeve.

[0021] In other features, the sleeve includes a plurality of knitted courses each having loops interlocking with loops of an adjacent knitted course of the plurality of knitted courses, and one or more of the knitted courses includes an inlay woven therein.

[0022] In other features, the sleeve includes a smart material.

[0023] In other features, the electrode calendering method further includes receiving a signal from a force or pressure sensor disposed between the sleeve and the roller, and controlling a temperature control device based on the signal to adjust a temperature of the sleeve.

[0024] Further areas of applicability of the present disclosure will become apparent from the detailed description, the claims and the drawings. The detailed description and specific examples are intended for purposes of illustration only and are not intended to limit the scope of the disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0025] The present disclosure will become more fully understood from the detailed description and the accompanying drawings, wherein:

[0026] FIG. 1 is a diagram of an electrode calendering system for reducing post-calendering electrode wrinkling, according to the present disclosure;

[0027] FIG. 2 is a diagram of an electrode and a roller of the electrode calendering system of FIG. 1, with sleeves wrapped around the roller according to the present disclosure;

[0028] FIG. 3 is a side view of the roller and one of the sleeves of FIG. 2, according to the present disclosure;

[0029] FIG. 4 is an isometric view of an example sleeve including a smart material, according to the present disclosure;

[0030] FIGS. 5-7 are examples of portions of the sleeve of FIG. 4, according to the present disclosure;

[0031] FIG. 8 is a graph plotting a stress-strain response of a smart material, according to the present disclosure;

[0032] FIG. 9 is a side view of a portion of an example sleeve having a single layer design with an integrated smart material, according to the present disclosure;

[0033] FIG. 10 is an isometric view of the sleeve of FIG. 9, according to the present disclosure;

[0034] FIG. 11 is a side view of a roller and the sleeve of FIG. 9, according to the present disclosure;

[0035] FIG. 12 is a side view of a portion of an example sleeve having a double layer design with an integrated smart material, according to the present disclosure;

[0036] FIG. 13 is a front view of a portion of an example sleeve having knitted bands of material, according to the present disclosure;

[0037] FIG. 14 is a front view of a portion of an example sleeve having knitted bands of material and inlays woven between loops of the knitted bands, according to the present disclosure;

[0038] FIG. 15 is a sectional view of one of the knitted bands of FIG. 14, according to the present disclosure;

[0039] FIG. 16 is an isometric view of a roller and the sleeve of FIG. 14, according to the present disclosure;

[0040] FIG. 17 is a side view of the roller and the sleeve of FIG. 16, according to the present disclosure;

[0041] FIG. 18 is a diagram of an electrode calendering system for reducing post-calendering electrode wrinkling, according to the present disclosure; and

[0042] FIG. 19 is a flowchart of an example process for reducing post-calendering electrode wrinkling, according to the present disclosure.

[0043] In the drawings, reference numbers may be reused to identify similar and/or identical elements.

DETAILED DESCRIPTION

[0044] During production of battery electrodes, defects can occur and/or exist in the electrode material, such as surface, internal and interface type defects. For example, significant wrinkling of battery cell electrodes may occur during an electrode calendering process. More specifically, battery cell electrodes are generally formed of foil having an interior coated region and uncoated bare edges on opposing sides of the interior coated region. In the calendering process, there is often a need for very high compressive force between rollers to produce higher electrode density. The compressive force is often applied on the interior coated region only. In some examples, this applied force configuration on the foil causes wrinkling and other failures of the uncoated foil (at the two edges).

[0045] The electrode calendering systems and methods according to the present disclosure provide solutions to reduce post-calendering electrode wrinkling by controlling the contact pressure between the uncoated edges of the electrode and calendering roller(s). In various embodiments, and as further explained below, the solutions include disposing sleeves about the outer curved surfaces of calendering rollers, along outer edge portions of the rollers. Each sleeve is passively and/or actively altered (e.g., without or with in-process sensing and feedback signals) to control or adjust a thickness of the sleeve and to adjust or modulate a contact pressure between the outer edge of the

electrode and the sleeve. In various embodiments, the thickness of the sleeve may be adjusted by a few microns or up to about 30 microns if desired.

[0046] Referring now to FIG. 1, a block diagram of an example electrode calendering system **100** is presented for reducing post-calendering electrode wrinkling. The electrode calendering system **100** of FIG. 1 and/or any of the other example systems and methods herein may be applicable in the manufacturing of battery cell electrodes (e.g., anode and cathode electrodes) for vehicle applications and/or any other suitable applications including electrodes.

[0047] As shown in FIG. 1, the electrode calendering system **100** generally includes multiple rollers **104, 106, 108, 110, 112, 114, 116, 118, 120, 122** and an electrode **102** (shown as a dashed line). In the example of FIG. 1, the rollers have various sizes for compressing and moving the electrode **102** through a cabinet **124** housing the rollers. The rollers may include idle rollers having fixed locations and dance rollers which may be adjusted (e.g., up and down in a vertical direction).

[0048] In the example of FIG. 1, the electrode **102** may be conventional. For example, the electrode **102** may be formed of foil or another suitable electrically conductive material. Additionally, the electrode **102** may have multiple regions. For instance, FIG. 2 depicts one example of the roller **114** and the electrode **102** of FIG. 1. As shown in FIG. 2, the electrode **102** generally includes an interior region (e.g., a coated region) **202** and outer edges (e.g., uncoated, bare edges) **204, 206** on opposing sides of the interior region **202**. In some examples, the outer edges **204, 206** may have the same width (e.g., as shown in FIG. 2) or different widths depending, for example, on the design of the electrode **102**. In such examples, both outer edges **204, 206** may be uncoated, bare edges. In other examples, only one of the outer edges **204, 206** may be an uncoated, bare edge while the other outer edge **204, 206** may be coated (e.g., similar to the interior region **202**).

[0049] Each roller **104, 106, 108, 110, 112, 114, 116, 118, 120, 122** of FIG. 1 is cylindrically shaped and defined by an outer surface for contacting (and sometimes compressing) the electrode **102**. For example, FIG. 3 depicts one example of the roller **114** of FIGS. 1-2. As shown in FIGS. 1-3, the roller **114** includes an outer surface **126** that contacts the electrode **102** when the electrode **102** is passed by the roller **114**. In this example, the outer surface **126** has edge portions **128, 130** at opposing sides (or ends) **132, 134** of the roller **114**. As shown, the edge portions **128, 130** of the roller **114** substantially align with the outer edges **204, 206** of the electrode **102** when the electrode contacts the roller **104**.

[0050] In various embodiments, one or more sleeves may be disposed about the outer surface of any one of the rollers **104, 106, 108, 110, 112, 114, 116, 118, 120, 122** of FIG. 1. In such examples, the sleeve(s) may be disposed around the roller outer surface in a circumferential direction. For example, and as shown in FIGS. 2-3, two sleeves **208, 210** are disposed about the outer surface **126** of the roller **114** along the edge portions **128, 130** of the roller **114**. In such examples, the sleeves **208, 210** are aligned with the outer edges **204, 206** of the electrode **102** when the electrode contacts the roller **104**, as shown in FIG. 2. While the example of FIGS. 2-3 is described with the roller **114** having two sleeves **208, 210** wrapped around its outer surface **126**, it should be appreciated that the roller **114** may include more or less sleeves desired depending on, for example, possible wrinkle location(s) of an electrode. Additionally, in various embodiments, other rollers may include one or more sleeves as disclosed herein. For example, the roller **116** of FIG. 1 may have one or more sleeves corresponding to the one or more sleeves on the roller **114**.

[0051] The sleeves **208, 210** of FIGS. 2-3 may include various characteristics. For example, and as further explained herein, the sleeves **208, 210** may include different types of material, different shapes, etc. to reduce post-calendering wrinkling of an electrode (e.g., the electrode **102**). This reduction in wrinkling is generally achieved by controlling the thicknesses of the sleeves **208, 210** to adjust contact pressures between the outer edges **204, 206** of the electrode **102** and the sleeves **208, 210**. For example, the stiffness of each sleeve **208, 210** may be adjusted (e.g., sometimes in real-time) to control the thickness of that sleeve **208, 210** when the sleeve contacts the corresponding outer edge **204, 206** of the electrode **102**. By adjusting the stiffness and therefore

thickness of the sleeves **208, 210**, the contact pressure between the electrode **102** and the sleeves **208, 210** wrapped around the roller **114** may be controlled to ensure a desired force is applied to the outer edges **204, 206** of the electrode **102**. In doing so, non-uniformities of the electrode **102** and/or the roller **114** (e.g., roller gaps, electrode coating variations, etc.) may be negated while a compressive force is applied to the outer edges **204, 206**, thereby reducing (and sometimes eliminating) wrinkling defects along the outer edges **204, 206** of the electrode **102**.

[0052] In various embodiments, the thickness of each sleeve **208, 210** may be passively or actively adjusted. For example, the stiffness of each sleeve **208, 210** may be passively modulated without in-process sensing and feedback signals. In such examples, the sleeves **208, 210** may include desirable components (e.g., materials, components, etc.), have desirable shapes, be formed in desirable manners, etc. to cause the thickness of the sleeves **208, 210** to change, as further explained below. In other examples, the stiffness of each sleeve **208, 210** may be actively modulated with in-process sensing and feedback signals to cause the thickness of the sleeves **208, 210** to change, as further explained below.

[0053] For example, the sleeves herein may be formed at least partially of smart materials. In such examples, the sleeves may include a matrix of non-active material and a smart material. This configuration of materials exhibits a stretchy characteristic, thereby allowing the sleeves to be stretched to slip on and off the roller. As one example, FIG. 4 depicts a sleeve **400** that may be employed as any one of the sleeves **208, 210** shown in FIGS. 2-3. In the example of FIG. 4, the sleeve **400** includes a base **402** defining an opening **404**. With this configuration, the base **402** (and more generally, the sleeve **400**) may be positioned about a roller (e.g., the roller **114** of FIGS. 1-3) such that the sleeve **400** contacts the outer surface of the roller and the roller extends through the opening **404**.

[0054] In the example of FIG. 4, the sleeve **400** includes at least one smart (or active) material to adaptively adjust the thickness/stiffness of the sleeve **400** and/or the contact pressure between the electrode and sleeve **400** without in-process sensing signals. For example, in one embodiment, the sleeve **400** may take the form of a woven belt including one or more yarns made of non-active/smart material (e.g., polyester, etc.) and an active/smart element made of a super-elastic shape memory alloy (SE-SMA). In such examples, the non-active yarns constitute a matrix that contains and locates the SE-SMA elements in the proper position within the belt. The matrix is stretchy (e.g., may undergo large elastically recoverable strains) due to, for example, the architecture of the weave, the properties of the non-active yarns, or both. The SE-SMA material exhibits large recoverable strain due to, for example, of a reversible, stress-driven solid-state phase transformation in the SE-SMA material. Additionally, the geometry of the SE-SMA element may also allow it to undergo recoverable deformations. As both components of the sleeve **400** (e.g., the matrix and the SE-SMA element) exhibit large recoverable strains, the sleeve **400** itself can undergo large (e.g., 2-10%) strain, which makes it possible for the sleeve **400** to be stretched to assemble it onto the roller or disassemble it from the roller.

[0055] In such examples, the shape memory alloy (SMA) may be a metal material with unique properties that allow the material to be trained to move (e.g., stretched, bent, etc.) and revert to its original shape. As an example, the SMA component may be based on a nickel-titanium alloy (e.g., nitinol, etc.). In other examples, the SMA component may be a copper-based alloy, a gold-based alloy, etc.

[0056] FIGS. 5-7 depict various sleeve portions **500, 502, 504, 600, 700** that may form a part of the sleeve **400** of FIG. 4. For example, in FIG. 5, the sleeve portion **500** includes a strand of yarn **512** (e.g., non-active/smart material such as, polyester, etc.) and an active/smart element **510** (e.g., formed of SE-SMA), the sleeve portion **502** includes two strands of yarn **522** and multiple active/smart elements **520**, and the sleeve portion **504** includes strands of yarn **532** and strands of an active/smart element **530**. Additionally, as shown in FIG. 6, the sleeve portion **604** includes strands of yarn **612** and strands of an active/smart element **610**. In FIG. 7, the sleeve portion **704**

includes strands of yarn **712** and strands of an active/smart element **710**.

[0057] In various embodiments, SE-SMA materials generally exhibit a plateau in their stress-strain responses. For example, FIG. **8** depicts a graph **800** plotting a stress-strain response of a SE-SMA material relative to the stress (y-axis) and the strain (x-axis). The graph **800** includes a line **802** representing the forward transformation (e.g., loading) of the SE-SMA material and a line **804** representing the reverse transformation (e.g., unloading) of the SE-SMA material. As shown, the SE-SMA material generally exhibits a plateau between the dashed vertical lines. In this region, the forward transformation line **802** and the reverse transformation line **804** have smaller positive slopes as compared to other regions along the lines **802**, **804**. With this approach, the material response can be utilized to create passive SE-SMA elements that exert a nearly constant-force (stress) over a strain range of about 2%-8%. As such, this design ensures constant pressure of the sleeve **400** on the calendaring roller (e.g., the roller **114**) that can easily accommodate roller gaps and/or electrode coating variations of up to 15 microns or more, which is more than enough for production purposes that generally have variations of 10 microns or less.

[0058] In various embodiments, the sleeves herein may be formed with integrated passive SE-SMA materials. In such examples, the stiffness of the sleeves may be greater than the stiffness of the sleeve **400** (e.g., including any one of the sleeve portions **500**, **502**, **504**, **600**, **700**) explained above. This may result in a better ability to control the applied stress at corners and edges of the electrode. For example, FIGS. **9-11** depict a single layer design of a sleeve **900** that may be employed as any one of the sleeves **208**, **210** shown in FIGS. **2-3** and wrapped about the roller **114**. In the example of FIGS. **9-11**, the sleeve **900** includes a support layer **902**, a smart material element **904** integrated into the support layer **902**, and an optional protective layer **910**. In such examples, the protective layer **910** may be disposed over the outer surface of the sleeve **900** such that it is interposed between the smart material element **904** and the electrode to improve the surface finish of the electrode. Additionally, the support layer **902** may be a belt formed of rubber, elastomer, leather, polymer, cork, etc. The smart material element **904** may include a SE-SMA material as explained above and in the form of a wire, strip, net, etc. In some examples, set screws or other suitable fastening devices may be employed to secure opposing ends of the smart material element **904** and the support layer **902** and to the roller.

[0059] In other examples, the sleeves herein may be formed with integrated passive SE-SMA materials in a double layer design. For example, FIG. **12** depicts a double layer design of a sleeve **1200** that may be employed as any one of the sleeves **208**, **210** shown in FIGS. **2-3**. In the example of FIG. **12**, the sleeve **1200** includes the support layer **902**, the smart material element **904**, and the optional protective layer **910** of FIGS. **9-11**. Additionally, the sleeve **1200** includes a support layer **1202** disposed along a bottom surface of the support layer **902** such that a portion of the smart material element **904** of the smart material is positioned between the support layers **902**, **904** as shown in FIG. **9**. In such examples, the sleeve **1200** may be formed in a manner so that the bottom surface of the support layer **1202** generally contacts the outer surface of the roller (e.g., the roller **114** of FIGS. **1-3**). In various embodiments, a coating (e.g., a high friction coating, etc.) may be disposed along the bottom surface of the support layer **1202** (e.g., the surface opposing the support layer **902**) and between the support layer **1202** and the outer surface of the roller when the sleeve **1200** is placed on the roller.

[0060] In various embodiments, the sleeves herein may be formed of knitted bands of material having radial compliance characteristics. In such examples, each knitted band may be formed of smart and/or non-smart materials. The band or belt of knitted material may be composed of a set of stitches a small field of the constituent stitches. A small field of the constituent stitches for a simple 'jersey' knit is depicted in FIG. **13** as a knit field **1300**. The knit field **1300** may be repeated in the circumferential and radial directions to construct any one of the sleeves **208**, **210** shown in FIGS. **2-3**. In the example of FIG. **13**, the knit field **1300** includes three courses **1302**, **1304**, **1306** of materials **1308**, **1310**, **1312**, respectively, knitted together. While the knit field **1300** of FIG. **13** is

shown as including three courses, it should be appreciated that other embodiments may include more or less courses if desired to create a band of the appropriate final geometry.

[0061] In such examples, standard knit materials may be constructed of yarn loops interlocking with an adjacent course of yarn above and/or below it. More specifically, and as shown in FIG. 13, the course **1302** has loops formed of the material **1308** that interlock with loops of the adjacent band **1304** formed of the material **1310**. Similarly, the course **1304** has loops formed of the material **1310** that interlock with loops of the adjacent course **1306** formed of the material **1312**. This configuration produces a uniform layer of material where the yarns overlap at the stitch loops.

[0062] The materials **1308**, **1310**, **1312** may be the same or different types, such as a mixture of smart materials and non-smart materials. For example, the courses **1302**, **1304**, **1306** may include alternating types of material, the courses **1302**, **1304** may have the same material and the course **1306** may have a different material, the courses **1302**, **1304**, **1306** may have the same material. In the example of FIG. 13, the materials **1308**, **1312** are non-smart materials while the material **1310** is a smart material (e.g., a SE-SMA material as explained above), thereby providing an alternating occurrence within the sleeve.

[0063] In some examples, other materials may be introduced into the knitted bands of material using additional machine operations, such as inlaying. In such examples, an inlay element may be woven between loops of a knitted band. For example, FIG. 14 depicts a knit field **1400** that may be employed to comprise any one of the sleeves **208**, **210** shown in FIGS. 2-3. In the example of FIG. 14, the knit field **1400** includes three courses **1402**, **1404**, **1406** of materials **1408**, **1410**, **1412**, respectively, knitted together in a similar manner as the courses **1302**, **1304**, **1306** of FIG. 13 described above. In the example of FIG. 14, the materials **1408**, **1410**, **1412** are formed of the same type of material, such a non-smart material or a smart material (e.g., are collocated in occurrence within the sleeve). While the knit field **1400** of FIG. 14 is shown as including three courses, it should be appreciated that knit field **1400** in other embodiments may include more or less courses if desired.

[0064] As shown in FIG. 14, each knitted course **1402**, **1404**, **1406** includes inlaid elements **1414**, **1416**, **1418** woven therein. For example, the knitted course **1402** includes the inlaid element **1414** woven between adjacent loops of the material **1408**. In such examples, the inlaid element **1414** passes in front of one loop and then behind the adjacent loop. The knitted courses **1404**, **1406** and the inlaid elements **1416**, **1418** are similarly configured. While the knitted courses **1402**, **1404**, **1406** are shown as having the inlaid elements **1414**, **1416**, **1418** woven between adjacent loops, it should be appreciated that the frequency of the front to back transitions of the inlaid elements **1414**, **1416**, **1418** with respect to the knitted courses **1402**, **1404**, **1406** may be different to control the radial compliance characteristics of the knit field **1400**.

[0065] In various embodiments, when the knitted courses **1402**, **1404**, **1406** are formed into a tubular structure for wrapping around a roller, the inlaid helix of yarn may develop a wave-spring like structure that will give radial compliance to the knit field **1400**. For example, FIG. 15 depicts a sectional view of the knitted course **1402** of FIG. 14 along the line **1430**. As shown, the inlaid element **1414** creates a sinusoidal shape (or more generally a wave like shape).

[0066] FIGS. 16-17 depict an assembly **1600** including the roller **114** of FIGS. 1-3 and the knit field **1400** of FIGS. 14-15. As described above relative to FIGS. 14-15, the knit field **1400** includes the knitted courses **1402**, **1404**, **1406** with the inlaid elements **1414**, **1416**, **1418** woven into the materials **1408**, **1410**, **1412**. As shown in FIGS. 16-17, the knit field **1400** has a sinusoidal shape in a circumferential direction relative to the roller **114**. In such examples, if fitted to the proper calendar roll diameter and knit with the appropriate materials, the tubular knit courses **1402**, **1404**, **1406** may act as a carrier for a helical wave spring structure created by the inlaid elements **1414**, **1416**, **1418**. The carrier knit imparts the bumpy pattern to the inlaid elements **1414**, **1416**, **1418**, creating a compliant surface on the roller **114**.

[0067] In various embodiments, the knitted courses **1302**, **1304**, **1306**, **1402**, **1404**, **1406** of FIGS.

13-17 may form a self-folding structure. For example, the knitted courses **1302, 1304, 1306, 1402, 1404, 1406** may be formed of basic jersey knit textiles, which have interconnecting loops joining adjacent courses of the fabric as explained above. The overlapping of the stitch loops may be consistent across a simple textile, but the stitches can loop behind (face) or in front (back) of the loops in the next course. In some embodiments, the jersey knit fabric has a tendency to curl due to this stitch architecture, but the direction of that curl is determined by how the loops overlap. For example, stitches that loop behind (face) may curl in one direction and stitches that loop in front (back) may curl in the opposite direction. As such, the knitted courses **1302, 1304, 1306, 1402, 1404, 1406** can be designed to create bowed surfaces that have dramatically increased compliance when placed on a cylindrical surface, such as the roller **114**. Further, the stitch length, spacing, and tightness can make this curling effect more or less pronounced; with denser, tighter knits typically curling more. Increasing the stiffness of the material also amplifies the curling, creating a “turn” in a textile with only a few courses of knit.

[0068] In other examples, the sleeves herein may be actively controlled with sensing and feedback signals. For example, actively controlled sleeves may be employed if passively designed sleeves are not sufficient. In various embodiments, the sleeves may be formed of smart materials, such as a SE-SMA material as explained above. In such examples, the plateau stress of the SE-SMA material can be controlled by adjusting a temperature of the smart material. For example, the plateau in a stress-strain response may rise by roughly 10.5 MPa for every degree rise in temperature of the smart material for nearly equ-atomic SMAs. Additionally, the same may apply in reverse. For example, the plateau stress or the transformation stress may decrease by 10.5 Mpa for every degree drop in temperature for the material. Limits may exist on how high or low of a temperature can be imposed on the material without compromising its performance. For example, a maximum or upper bound on temperature may be lower than the recrystallization temperature for the SE-SMA material. The lower bound on temperature is usually less critical. This temperature dependence can be used to actively modulate the gripping pressure on an electrode (e.g., the electrode **102**) by using directed, non-contact, heating/cooling sources to control the local temperature of the smart material in the region where the electrode tabs are gripped by the sleeves.

[0069] For example, FIG. **18** depicts an example electrode calendering system **1800** for reducing post-calendering electrode wrinkling. The electrode calendering system **1800** of FIG. **18** is substantially similar to the electrode calendering system **100** of FIG. **1** but includes control components for actively controlling sleeves. For example, the system **1800** includes the multiple rollers (of which only the rollers **114, 116** are labeled for clarity) and the electrode **102** of FIG. **1**. Additionally, the system **1800** includes a control module **1802**, sensors **1804, 1806**, temperature control devices **1808, 1810**, and an optional optical sensor **1812**. In various embodiments, each sensor **1804, 1806** may include a pressure sensor, a force sensor and/or any other suitable sensor. As shown, the sensors **1804, 1806**, the temperature control devices **1808, 1810**, and the optical sensor **1812** are in communication with the control module **1802**.

[0070] In the example of FIG. **18**, the sensors **1804, 1806** are generally disposed between sleeves and rollers to measure (e.g., in real-time) the contact pressure distribution to feedback to the control module **1802**. For example, the rollers **114, 116** may include one or more sleeves (e.g., any one of the sleeves described herein), which are not shown for clarity. In such examples, the sensor **1804** is disposed between a sleeve wrapped around the roller **114** and the roller **114**, and the sensor **1806** is disposed between a sleeve wrapped around the roller **116** and the roller **116**. In some examples, the sensors **1804, 1806** may include a sensor array.

[0071] In various embodiments, the temperature control devices **1808, 1810** are adjacent to the sleeves wrapped around the rollers **114, 116**. In such examples, one or both temperature control devices **1808, 1810** may be controlled (e.g., individually or collectively) to heat or cool one or more of the sleeves. In some examples, the temperature control devices **1808, 1810** may be heating/cooling sources (e.g., HVAC modules) having one or more fans, heaters, air conditioners,

infrared (IR) lights, etc. for heating or cooling the sleeves without contacting the sleeves.

[0072] In FIG. **18**, the optical sensor **1812** is generally disposed downstream of the rollers **114**, **116**. In such examples, the optical sensor **1812** may include one or more cameras or other suitable devices for capturing images of the electrode **102** downstream of the rollers **114**, **116**. In such manner, the optical sensor **1812** may capture imaginary of the post-calendering electrode **102**. [0073] In various embodiments, the control module **1802** receives one or more signals from one or both of the sensors **1804**, **1806**. In such examples, the signals may be indicative of a force or pressure value between the corresponding sleeve and roller. Then, based on the received signals, the control module **1802** may control one or both temperature control devices **1808**, **1810** to adjust a temperature of the sleeve, thereby adjusting the pressure between the electrode **102** and the sleeve.

[0074] In some examples, the control module **1802** may receive one or more signals from the optical sensor **1812**. In such examples, the signals provide data to indicate whether the post-calendering electrode **102** has wrinkles or not. In such examples, the control module **1802** may control one or both temperature control devices **1808**, **1810** based on the signals from the optical sensor **1812** to adjust a temperature of the sleeve, thereby adjusting the pressure between the electrode **102** and the sleeve. For instance, the control module **1802** may adjust control of the temperature control devices **1808**, **1810** based on the data from the optical sensor **1812** and the sensors **1804**, **1806**. In some examples, the control module **1802** may initiate control based on the signal feedback in response to data from the optical sensor **1812** indicating wrinkling of the electrode **102**.

[0075] FIG. **19** illustrates an example process **1900** for reducing post-calendering electrode wrinkling. While FIG. **19** is shown and described as including specific steps, it should be appreciated that the process **1900** is an example variation that may be implemented and in other embodiments the process **1600** and/or other example processes may include different steps, more or less steps, etc. Additionally, although the process **1900** is described in relation to the system **1800** of FIG. **18**, the process **1900** may be employable by any suitable system.

[0076] As shown, the process **1900** begins at **1902** where control determines whether an electrode calendering process has been activated. For example, control may sense (e.g., with a motion sensor, etc.) whether an electrode (e.g., the electrode **102**) is received by machinery including rollers. In other examples, a user may select an input once the electrode is received. If no at **1902**, control returns to **1902**. Otherwise, if yes at **1902**, control proceeds to **1904**.

[0077] At **1904**, the control module **1802** receives data from one or more post calendering sensors. For example, the control module **1802** may receive signals from the optical sensor **1812** indicating whether the electrode has wrinkles or not after passing through one or more rollers. Control then proceeds to **1906**, where the control module **1802** determines whether the electrode has defects (e.g., wrinkles along edges of the electrodes) as explained herein. If the control module **1802** determines that defects are present, control proceeds to **1908**. Otherwise, if the control module **1802** determines that defects are not present, control may return to **1902** as shown or may end.

[0078] At **1908**, the control module **1802** receives one or more force or pressure values. For example, and as explained above, the control module **1802** may receive one or more force or pressure signals from one or both of the sensors **1804**, **1806** located between roller sleeve(s) and the rollers. In such examples, the force or pressure signals may be indicative of a contact pressure distribution. Control then proceeds to **1910**.

[0079] At **1910**, the control module **1802** determines a temperature adjustment for the sleeve(s) based on the force or pressure value(s). For example, and as explained above, the plateau stress of the smart material in the sleeve(s) is controllable by adjusting the temperature of the material. For example, to increase the plateau stress (e.g., the applied force) of the smart material in the sleeve(s), the temperature of the material may be increased. Conversely, to decrease the plateau stress of the smart material in the sleeve(s), the temperature of the material may be decreased. In

such examples, the temperature adjustment (e.g., increase or decrease of the temperature) may be determined based on the received force or pressure values. Control then proceeds to **1912**.

[0080] At **1912**, the control module **1802** generates and transmits one or more control signals to one or more temperature control devices. For example, the control module **1802** may generate and transmit the control signals to the temperature control devices **1808**, **1810** of FIG. **18** based on the determined temperature adjustment. In such examples, the control module **1802** may transmit a control signal to initiate cooling or heating of the sleeve(s) with the temperature control devices **1808**, **1810**. In doing so, the thickness of the sleeve(s) may be controlled to adjust a pressure between the electrode and the sleeve(s), as explained herein. Control then may end as shown in FIG. **19** or return to another suitable step, such as step **1908** if desired.

[0081] The foregoing description is merely illustrative in nature and is in no way intended to limit the disclosure, its application, or uses. The broad teachings of the disclosure can be implemented in a variety of forms. Therefore, while this disclosure includes particular examples, the true scope of the disclosure should not be so limited since other modifications will become apparent upon a study of the drawings, the specification, and the following claims. It should be understood that one or more steps within a method may be executed in different order (or concurrently) without altering the principles of the present disclosure. Further, although each of the embodiments is described above as having certain features, any one or more of those features described with respect to any embodiment of the disclosure can be implemented in and/or combined with features of any of the other embodiments, even if that combination is not explicitly described. In other words, the described embodiments are not mutually exclusive, and permutations of one or more embodiments with one another remain within the scope of this disclosure.

[0082] Spatial and functional relationships between elements (for example, between modules, circuit elements, semiconductor layers, etc.) are described using various terms, including “connected,” “engaged,” “coupled,” “adjacent,” “next to,” “on top of,” “above,” “below,” and “disposed.” Unless explicitly described as being “direct,” when a relationship between first and second elements is described in the above disclosure, that relationship can be a direct relationship where no other intervening elements are present between the first and second elements, but can also be an indirect relationship where one or more intervening elements are present (either spatially or functionally) between the first and second elements. As used herein, the phrase at least one of A, B, and C should be construed to mean a logical (A OR B OR C), using a non-exclusive logical OR, and should not be construed to mean “at least one of A, at least one of B, and at least one of C.”

[0083] In the figures, the direction of an arrow, as indicated by the arrowhead, generally demonstrates the flow of information (such as data or instructions) that is of interest to the illustration. For example, when element A and element B exchange a variety of information but information transmitted from element A to element B is relevant to the illustration, the arrow may point from element A to element B. This unidirectional arrow does not imply that no other information is transmitted from element B to element A. Further, for information sent from element A to element B, element B may send requests for, or receipt acknowledgements of, the information to element A.

[0084] In this application, including the definitions below, the term “module” or the term “controller” may be replaced with the term “circuit.” The term “module” may refer to, be part of, or include: an Application Specific Integrated Circuit (ASIC); a digital, analog, or mixed analog/digital discrete circuit; a digital, analog, or mixed analog/digital integrated circuit; a combinational logic circuit; a field programmable gate array (FPGA); a processor circuit (shared, dedicated, or group) that executes code; a memory circuit (shared, dedicated, or group) that stores code executed by the processor circuit; other suitable hardware components that provide the described functionality; or a combination of some or all of the above, such as in a system-on-chip.

[0085] The module may include one or more interface circuits. In some examples, the interface circuits may include wired or wireless interfaces that are connected to a local area network (LAN),

the Internet, a wide area network (WAN), or combinations thereof. The functionality of any given module of the present disclosure may be distributed among multiple modules that are connected via interface circuits. For example, multiple modules may allow load balancing. In a further example, a server (also known as remote, or cloud) module may accomplish some functionality on behalf of a client module.

[0086] The term code, as used above, may include software, firmware, and/or microcode, and may refer to programs, routines, functions, classes, data structures, and/or objects. The term shared processor circuit encompasses a single processor circuit that executes some or all code from multiple modules. The term group processor circuit encompasses a processor circuit that, in combination with additional processor circuits, executes some or all code from one or more modules. References to multiple processor circuits encompass multiple processor circuits on discrete dies, multiple processor circuits on a single die, multiple cores of a single processor circuit, multiple threads of a single processor circuit, or a combination of the above. The term shared memory circuit encompasses a single memory circuit that stores some or all code from multiple modules. The term group memory circuit encompasses a memory circuit that, in combination with additional memories, stores some or all code from one or more modules.

[0087] The term memory circuit is a subset of the term computer-readable medium. The term computer-readable medium, as used herein, does not encompass transitory electrical or electromagnetic signals propagating through a medium (such as on a carrier wave); the term computer-readable medium may therefore be considered tangible and non-transitory. Non-limiting examples of a non-transitory, tangible computer-readable medium are nonvolatile memory circuits (such as a flash memory circuit, an erasable programmable read-only memory circuit, or a mask read-only memory circuit), volatile memory circuits (such as a static random access memory circuit or a dynamic random access memory circuit), magnetic storage media (such as an analog or digital magnetic tape or a hard disk drive), and optical storage media (such as a CD, a DVD, or a Blu-ray Disc).

[0088] The apparatuses and methods described in this application may be partially or fully implemented by a special purpose computer created by configuring a general purpose computer to execute one or more particular functions embodied in computer programs. The functional blocks, flowchart components, and other elements described above serve as software specifications, which can be translated into the computer programs by the routine work of a skilled technician or programmer.

[0089] The computer programs include processor-executable instructions that are stored on at least one non-transitory, tangible computer-readable medium. The computer programs may also include or rely on stored data. The computer programs may encompass a basic input/output system (BIOS) that interacts with hardware of the special purpose computer, device drivers that interact with particular devices of the special purpose computer, one or more operating systems, user applications, background services, background applications, etc.

[0090] The computer programs may include: (i) descriptive text to be parsed, such as HTML (hypertext markup language), XML (extensible markup language), or JSON (JavaScript Object Notation) (ii) assembly code, (iii) object code generated from source code by a compiler, (iv) source code for execution by an interpreter, (v) source code for compilation and execution by a just-in-time compiler, etc. As examples only, source code may be written using syntax from languages including C, C++, C#, Objective-C, Swift, Haskell, Go, SQL, R, Lisp, Java®, Fortran, Perl, Pascal, Curl, OCaml, JavaScript®, HTML5 (Hypertext Markup Language 5th revision), Ada, ASP (Active Server Pages), PHP (PHP: Hypertext Preprocessor), Scala, Eiffel, Smalltalk, Erlang, Ruby, Flash®, Visual Basic®, Lua, MATLAB, SIMULINK, and Python®.

Claims

- 1.** An electrode calendering system for reducing post-calendering electrode wrinkling, the electrode calendering system comprising: at least one roller configured to compress an electrode, the roller defined by an outer surface configured to contact the electrode, the outer surface having a first edge portion at a first side of the roller and a second edge portion at a second side of the roller opposing the first side, the first edge portion and the second edge portion of the roller aligning with first and second outer edges of the electrode when the electrode contacts the roller; and a sleeve disposed about the outer surface of the roller along the first edge portion, the sleeve having a controllable thickness to adjust a pressure between the first outer edge of the electrode and the sleeve.
- 2.** The electrode calendering system of claim 1, wherein: the sleeve is a first sleeve; the electrode calendering system further comprises a second sleeve disposed on the outer surface of the roller along the second edge portion; and the second sleeve has a controllable thickness to adjust a pressure between the second outer edge of the electrode and the second sleeve.
- 3.** The electrode calendering system of claim 1, wherein the thickness of the sleeve adjusts passively to modulate the pressure between the first outer edge of the electrode and the sleeve.
- 4.** The electrode calendering system of claim 1, wherein the sleeve includes a plurality of knitted courses each having loops interlocking with loops of an adjacent knitted band of the plurality of knitted bands.
- 5.** The electrode calendering system of claim 4, wherein one or more of the knitted courses includes an inlay element woven therein.
- 6.** The electrode calendering system of claim 4, wherein the sleeve includes multiple inlaid elements having a sinusoidal shape in a circumferential direction relative to the roller.
- 7.** The electrode calendering system of claim 4, wherein: the plurality of knitted courses includes a first set of courses formed of a smart material and a second set of courses formed of non-smart materials; and the first set of courses and the second set of courses are collocated or alternated in occurrence within the sleeve.
- 8.** The electrode calendering system of claim 1, wherein the sleeve includes a smart material element.
- 9.** The electrode calendering system of claim 8, wherein the smart material element includes a super-elastic shape memory alloy (SE-SMA).
- 10.** The electrode calendering system of claim 8, wherein the sleeve includes a support layer and the smart material element integrated into the support layer.
- 11.** The electrode calendering system of claim 10, wherein: the support layer is a first support layer; and the sleeve includes a second support layer disposed along a surface of the first support layer such that a portion of the layer of the smart material is positioned between the first support layer and the second support layer.
- 12.** The electrode calendering system of claim 1, wherein: the electrode calendering system further comprises at least one optical sensor disposed downstream of the roller; the control module in communication with the optical sensor; and the control module is configured to receive a signal from the optical sensor and control the temperature control device based on the signal from the optical sensor to adjust a temperature of the sleeve, thereby adjusting the pressure between the first outer edge of the electrode and the sleeve.
- 13.** The electrode calendering system of claim 1, wherein the thickness of the sleeve is actively controlled to adjust the pressure between the first outer edge of the electrode and the sleeve.
- 14.** The electrode calendering system of claim 1, further comprising: a force or pressure sensor disposed between the sleeve and the roller; a temperature control device adjacent to the sleeve; and a control module in communication with the force or pressure sensor and the temperature control device, the control module configured to receive a signal from the force or pressure sensor and control the temperature control device based on the signal to adjust a temperature of the sleeve, thereby adjusting the pressure between the first outer edge of the electrode and the sleeve.

- 15.** An electrode calendering method for reducing post-calendering electrode wrinkling, the electrode calendering method comprising: receiving an electrode at a roller, the roller defined by an outer surface configured to contact the electrode, the outer surface having a first edge portion at a first side of the roller and a second edge portion at a second side of the roller opposing the first side, the first edge portion and the second edge portion of the roller aligning with first and second outer edges of the electrode when the electrode contacts the roller; and controlling a thickness of a sleeve disposed about the outer surface of the roller along the first edge portion, to adjust a pressure between the first outer edge of the electrode and the sleeve.
- 16.** The electrode calendering method of claim 15, wherein controlling the thickness of the sleeve includes passively adjusting the thickness of the sleeve to modulate the pressure between the first outer edge of the electrode and the sleeve.
- 17.** The electrode calendering method of claim 15, wherein controlling the thickness of the sleeve includes actively controlling thickness of the sleeve to adjust the pressure between the first outer edge of the electrode and the sleeve.
- 18.** The electrode calendering method of claim 15, wherein: the sleeve includes a plurality of knitted courses each having loops interlocking with loops of an adjacent knitted course of the plurality of knitted courses; and one or more of the knitted courses includes an inlay woven therein.
- 19.** The electrode calendering method of claim 15, wherein the sleeve includes a smart material.
- 20.** The electrode calendering method of claim 19, further comprising: receiving a signal from a force or pressure sensor disposed between the sleeve and the roller; and controlling a temperature control device based on the signal to adjust a temperature of the sleeve.
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